MCSC-103 • 4 CREDITS

Advanced Computer Networks and Communication

Connect physical transmission, network protocols, management, mobility, and security into one operational view.

STUDY BLOCKS	FORMAT	SCOPE
9	Definitions Equations Worked examples	Supplied syllabus only

How to use this handbook

For every study block: define the terms, reproduce the relation and its assumptions, solve the example without looking, and explain when the method is unsuitable.

Scope note: This handbook expands only topics named in the supplied syllabus. Confirm examination guidance, prescribed texts, and institutional policy with the responsible instructor.

Course map

1. Telecommunications foundations

Telecommunications history, Conducted media, Radiated media, Transmission characteristics

2. Coding, modulation and errors

Coding techniques, Modulation techniques, Error characteristics, Error detection, Error correction

3. Networks, standards and layered design

LAN and WAN applications, hardware, and software, OSI model, Industry standards, Topologies, Protocols, Internetworking devices

4. Applications and system selection

Information system applications, Selection of telecommunications and networking systems, Communications management

5. Security, signatures and authentication

Security and recovery, Communication security, Digital signatures, Authentication protocols

6. Wireless mobility

WLAN, Mobile IP

7. SNMP models and proxying

SNMP v1 and v2 organizational model, System overview, Information model, Communication model, Functional model, SNMP proxy server

8. Broadband, ATM and network management

Broadband networks and services, ATM technology, Configuration management, Fault management, Performance management, Network management tools, Network statistics measurement systems

9. Web and XML management plus laboratory

Web-based management, XML-based network management, Programming protocols using C/C++ and MATLAB, Theory practice

Study block 01

Telecommunications foundations

Telecommunications history	Conducted media
Radiated media	Transmission characteristics

Definition and idea

Telecommunication moves information over distance. Conducted media guides signals through copper or fiber; radiated media uses electromagnetic propagation. Bandwidth, attenuation, noise, delay, and signal-to-noise ratio shape achievable communication.

```
Shannon capacity: C = B log2(1 + S/N)
```

Worked example

Increasing bandwidth raises theoretical capacity, but noise and attenuation still limit reliable transmission. Fiber offers high bandwidth and low electromagnetic interference; radio supports mobility but shares spectrum.

> **MUST UNDERSTAND**
> Compare a medium by bandwidth, reach, interference, mobility, cost, and security exposure.

Study block 02

Coding, modulation and errors

Coding techniques	Modulation techniques
Error characteristics	Error detection
Error correction	

Definition and idea

Line and channel coding map information into transmitted symbols. Modulation varies a carrier. Detection adds redundancy to identify corruption; correction adds enough structured redundancy to recover some errors.

```
Hamming distance d_min detects up to d_min-1 errors and corrects floor((d_min-1)/2)
```

Worked example

A parity bit detects any odd number of flipped bits but cannot identify which bit changed. A CRC treats the bit string as a polynomial and checks a transmitted remainder.

MUST UNDERSTAND
Distinguish source coding, channel coding, line coding, and modulation.

Study block 03

Networks, standards and layered design

LAN and WAN applications, hardware, and software	OSI model
Industry standards	Topologies
Protocols	Internetworking devices

Definition and idea

Layering separates concerns through services and protocols. The OSI model organizes communication into seven conceptual layers. Hubs, switches, routers, and gateways act at different scopes; standards allow independently built systems to interoperate.

```
End-to-end delay = processing + queueing + transmission + propagation
```

Worked example

A switch forwards frames within a LAN using link-layer addresses; a router forwards packets between networks using network-layer addresses.

MUST UNDERSTAND
For any device or protocol, name its layer, inputs, outputs, and failure scope.

Study block 04

Applications and system selection

Information system applications	Selection of telecommunications and networking systems
Communications management	

Definition and idea

Network design translates application requirements into topology, capacity, redundancy, manageability, security, and cost choices. Selection should follow measured requirements rather than brand preference.

```
Availability = uptime / total time
```

Worked example

A video service prioritizes throughput and jitter control; a transaction system prioritizes correctness, latency, availability, and secure recovery. A requirements matrix makes trade-offs visible.

MUST UNDERSTAND
Record functional needs, service-level targets, risks, and total cost before selecting a system.

Study block 05

Security, signatures and authentication

Security and recovery	Communication security
Digital signatures	Authentication protocols

Definition and idea

Confidentiality prevents unauthorized disclosure, integrity detects alteration, and availability preserves service. Digital signatures bind a private-key operation to data; authentication protocols establish identity or possession of credentials.

```
signature = Sign(private_key, Hash(message))
```

Worked example

A verifier recomputes the message hash and verifies the signature with the public key. A nonce prevents an old authentication response from being replayed as fresh proof.

> **MUST UNDERSTAND**
> Separate encryption from signatures and authentication from authorization.

Study block 06

Wireless mobility

WLAN	Mobile IP

Definition and idea

A WLAN connects devices over radio through access points or peer modes. Mobility introduces association, roaming, interference, and security concerns. Mobile IP preserves reachability as a node changes its point of attachment.

```
Effective wireless throughput < nominal link rate
```

Worked example

A roaming device may reassociate with a new access point while higher layers try to preserve sessions; congestion and interference can dominate performance even with a strong signal.

MUST UNDERSTAND
Analyze mobility at link, network, transport, and application layers.

Study block 07

SNMP models and proxying

SNMP v1 and v2 organizational model	System overview
Information model	Communication model
Functional model	SNMP proxy server

Definition and idea

SNMP managers query or configure agents using managed objects defined in MIB modules. The information model names typed objects; the communication model exchanges protocol data units; the functional model covers monitoring and control. A proxy represents devices that cannot communicate directly with the manager.

```
managed object identifier = hierarchical OID
```

Worked example

A manager polls an interface counter, compares successive values, and estimates utilization. Traps or notifications report selected events without waiting for a poll.

MUST UNDERSTAND
Explain manager, agent, MIB, OID, polling, notification, and proxy roles.

Study block 08

Broadband, ATM and network management

Broadband networks and services	ATM technology
Configuration management	Fault management
Performance management	Network management tools
Network statistics measurement systems	

Definition and idea

ATM transports fixed-size cells through virtual connections. Operational management keeps inventory and configuration consistent, detects and isolates faults, and measures performance indicators such as latency, loss, throughput, and utilization.

```
utilization = observed traffic / link capacity
```

Worked example

A monitoring tool correlates rising interface errors with packet loss, while configuration history reveals whether a recent change preceded the fault.

> **MUST UNDERSTAND**
> For each metric, define its unit, collection interval, baseline, and alert threshold.

Study block 09

Web and XML management plus laboratory

Web-based management	XML-based network management
Programming protocols using C/C++ and MATLAB	Theory practice

Definition and idea

Web-based management exposes operational views through browser-accessible interfaces. XML represents structured management data. Laboratory work connects protocol theory to parsers, sockets, traffic experiments, and numerical analysis.

```
goodput = useful application bits / elapsed time
```

Worked example

A lab can implement a client-server message exchange, inject controlled loss, measure round-trip time, and plot results in MATLAB while documenting protocol states.

> **MUST UNDERSTAND**
> Every lab report should state topology, assumptions, procedure, raw observations, calculations, and limitations.

Revision checklist

Use this final pass before an assessment or interview:

- ■ I can define every listed syllabus term in plain language.
- ■ I can reproduce the central equations and state their assumptions.
- ■ I can complete each worked example and create one variation.
- ■ I can compare closely related methods and justify a choice.
- ■ I can name limitations, failure modes, and ethical or operational concerns where relevant.